

Who bears the costs of inflation?

Euro area households and the 2021–2022 shock

Filippo Pallotti, Gonzalo Paz-Pardo, Jiri Slacalek,
Oreste Tristani, Giovanni L. Violante

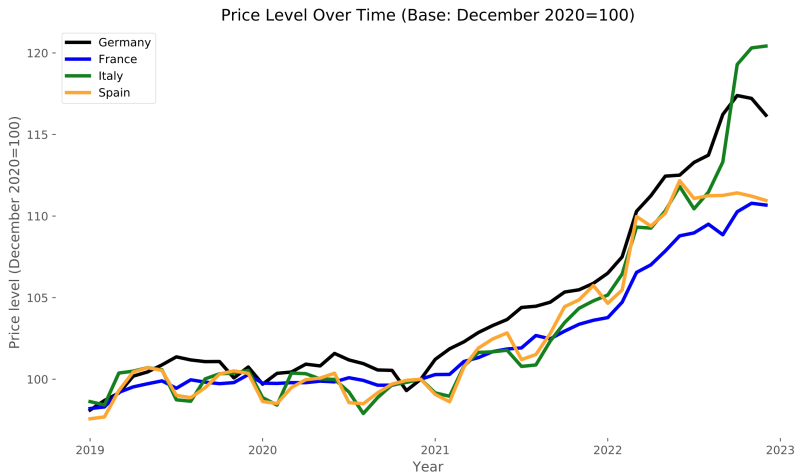
Presenter: Bill Shu

BGGMP Reading Group

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The Inflation Episode of 2021-2022

- Marked and unexpected increase in inflation driven by energy and food prices (10% in France and Spain, 16% in Germany, 20% in Italy)



Overview

- This paper: estimate the heterogeneous effects of surprise inflation across individual households in the recent inflation episode of 2021-2022
 - ▶ An event study focusing on the four largest euro area countries: Germany, France, Italy, and Spain Literature Review
- Measurement guided by theory:
 1. A tractable dynamic framework that illustrates various mechanisms through which inflation can affect household welfare
 2. Combine several micro and macro data sources to examine the relevance of all these channels across age and income distribution for each country

Key results

- Aggregate impact of this inflation episode is large: 3-4 % of biennial disposable income in Germany, France, and Spain; 8% in Italy
 - ▶ Much larger than standard estimates in a typical business cycle
- Older households suffered the most in welfare
 - ▶ Driven by their positive, large nominal share of financial portfolios, and large spending on heating and food
- No significant differences across consumption quintiles (proxy for permanent income) due to rent stickiness
- The model distinguishes between four channels of surprise inflation:
 - ▶ The direct effect of raw price changes dominates
 - ▶ Unconventional fiscal policies also played a key role in offsetting the heterogeneous direct effects

Roadmap

1. Framework

- 1.1. Basic Setup
- 1.2. Household's Problem
- 1.3. Welfare Decomposition

2. Empirical Implementation

3. Results

- 3.1. Heterogeneity in Inflation Rates
- 3.2. Welfare Decomposition

Environment

Basic Setting

- Overlapping-generations of households who live for two periods
- No aggregate uncertainty
- Before $t = 0$: economy in a steady state (ss); inflation normalized to 0

Unanticipated inflationary shock

- An aggregate disturbance dz_0 that induces a permanent jump in the aggregate price level at $t = 0$ (short run)
 - ▶ Heterogeneous real effects due to relative price changes, as well as different nominal income and positions in nominal assets
- $t = 1$ (long run): all effects are reabsorbed and price level constant
 - ▶ All nominal rigidities vanish in the long run, relative prices realigned
 - ▶ The young cohort lives through the long-run adjustment, while the old cohort does not

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Price Indices

- Consumption categories $j = 1, \dots, J$ with price levels $\mathcal{P}_{jt}$
- At time t , household i consumes the bundle $\{c_{i,jt}\}_{j=1,\dots,J}$, represented by a consumption aggregator c_{it} (HH i 's basket)
- P_{it} : the individual-level price index faced by household i (deflator for c_{it}):

$$c_{it}P_{it} = \sum_{j=1}^J c_{i,jt}\mathcal{P}_{jt}$$

- $\bar{P}_t$: the average price index of the economy
 - ▶ Similarly defined, using the economy's average consumption bundle $\bar{c}_t$ and nationwide average good j consumption $\bar{c}_{jt}$.
- Use average consumption bundle $\bar{c}_t$ as the numeraire of the economy, and its deflator $\bar{P}_t$ normalized to 1 in the initial steady state

Price Indices

- Consider the pre-tax price / raw price $\mathcal{P}_{jt}^*$:

$$\mathcal{P}_{jt} = \mathcal{P}_{jt}^*(1 + \tau_{jt})$$

where τ_{jt} is a wedge capturing good-specific taxes (if positive) or subsidies / price control measures (if negative)

- ▶ Can capture government interventions in the energy sector to mitigate hike in unit prices
- Now using pre-tax prices $\mathcal{P}_{jt}^*$, we can also define:
 - ▶ Pre-tax individual price indices $\mathcal{P}_{it}^*$
 - ▶ Pre-tax aggregate price index $\bar{\mathcal{P}}_t^*$

Log Deviations from Pre-shock Steady-State

- Up to the first order, change in household specific price index at $t = 0$:

$$d \log P_{i0} = \sum_{j=1}^J xsh_{ij,ss} \cdot d \log P_{j0}$$

where $xsh_{ij,ss} = \frac{c_{ij,ss} P_{j,ss}}{\sum_{j=1}^J c_{ij,ss} P_{j,ss}}$ is the nominal expenditure share on good j in steady state

- We can similarly get change in pre-tax individual index $d \log P_{i0}^*$
 - Can separate pre- and post-tax prices to identify the role of ad-hoc government interventions (e.g. $d \log P_{it} - d \log P_{it}^*$)

Log Deviations from Pre-shock Steady-State

- Similarly, the change in the aggregate price index is

$$d \log \bar{P}_0 = \sum_{j=1}^J \overline{xs}h_{j,ss} \cdot d \log \mathcal{P}_{j0}$$

where $\overline{xs}h_{j,ss}$ is the aggregate share spent on good J before the shock hits

- Can separate the average effect and individual deviations from the average to highlight the heterogeneous consequences of the shock (e.g. $d \log P_{it} - d \log \bar{P}_t$)

Budget Constraint

$$c_{it}P_{it} = W_{it} - T_{it} + B_{i,S,t} + B_{i,L,t} + \sum_{k=1}^K (Q_{kt} + D_{kt})a_{i,kt} \\ - Q_{S,t}B_{i,S,t+1} - Q_{L,t}(B_{i,L,t+1} - \delta B_{i,L,t}) - \sum_{k=1}^K Q_{kt}a_{i,kt+1}$$

- Earn nominal labor income W_{it}
- Pay nominal tax liability (net of transfers) T_{it} to the government
 - ▶ $T_{it} = T_{it}^{AUT} + T_{it}^{HOC}$ is the sum of "automatic stabilizers" already in place and ad-hoc fiscal responses by the government at $t = 0$
- Holdings of real assets (stocks, housing, etc.) $a_{i,kt}$, $k = 1, \dots, K$
 - ▶ Asset k trade at price Q_{kt} and pay a nominal dividend D_{kt}

Budget Constraint

- One-period (short-term) nominal bond: B_{i,S_t} with price Q_{S_t}
- Long-term bonds: following the conventional approach in the sovereign default literature (Arellano and Ramanarayanan, 2012)
 - ▶ Perpetuity contract with nominal coupon payments that decay at rate $\delta < 1$
 - ▶ Let Q_{L_t} be the price of new issuance and ℓ_{i,L_t} be purchase at time t
 - ▶ Nominal face value B_{it} : total payments due in period t on all purchases of past issuances:

$$B_{it} = \sum_{n=1}^t \delta^{n-1} \ell_{i,t-n} \implies B_{i,L_t+1} = \delta B_{i,L_t} + \ell_{it}$$

- B_{i,S_t} and B_{i,L_t} can be negative; households only face natural credit limits, no liquidity constraints

Young Household's Problem

Taking good prices, wages, net taxes, dividend policies, and all asset prices as given, for time $t = 0, 1$, the household chooses

- J consumption goods $\{c_{i,jt}\}_{j=1}^J$
- Holdings of real and nominal assets $\{a_{i,kt+1}\}_{k=1}^K, B_{i,St+1}, B_{i,Lt+1}$.

to maximize lifetime utility

$$V_i = u_i(c_{i0}) + \beta_i u_i(c_{i1})$$

subject to the budget constraints at $t = 0, 1$:

$$c_{it}P_{it} = W_{it} - T_{it} + B_{i,St} + B_{i,Lt} + \sum_{k=1}^K (Q_{kt} + D_{kt})a_{i,kt} \\ - Q_{St}B_{i,St+1} - Q_{Lt}(\underbrace{B_{i,Lt+1} - \delta B_{i,Lt}}_{\ell_{it}}) - \sum_{k=1}^K Q_{kt}a_{i,kt+1}$$

Nature of the Shock

- Interested in the impact on households' welfare of an exogenous shock dz_0 at time $t = 0$ that causes an increase in the price level equal to $d \log \bar{P}_0$
- **Assumption 1: the shock is unanticipated**
 - ▶ Burst of inflation is a surprise: not incorporated in prices and nominal variables at time $t = 0$
 - ▶ Allow the shock to affect relative prices in the short-run
- **Assumption 2: The inflation shock is temporary**
 - ▶ After the initial unexpected jump $d \log \bar{P}_0$, in the long run (from $t = 1$ onward), the aggregate price index remains and inflation returns to steady state (normalized to 0):

$$\frac{d \log \bar{P}_1}{dz_0} = \frac{d \log \bar{P}_0}{dz_0}$$

Assumption 4

Nature of the Shock

- **Assumption 3: In the long-run, the shock is neutral in the aggregate and across the distribution (individual inflation = aggregate inflation)**

- ▶ Wages, net taxes, dividends, real asset prices adjust one-to-one with the new aggregate price level:

$$\frac{d \log W_{i1}}{dz_0} = \frac{d \log T_{i1}}{dz_0} = \frac{d \log D_{k1}}{dz_0} = \frac{d \log Q_{k1}}{dz_0} = \frac{d \log \bar{P}_1}{dz_0}$$

- ▶ Expected inflation returns to steady state, so nominal bond prices return to initial values

$$\frac{d \log Q_{L1}}{dz_0} = \frac{d \log Q_{S1}}{dz_0} = 0$$

- ▶ This long-run realignment occurs also with respect to individual price levels:

$$\frac{d \log P_{i1}}{dz_0} = \frac{d \log \bar{P}_1}{dz_0}$$

Welfare Metric

- $\mathcal{L}_i$: the Lagrangean for household i in the cohort born at time $t = 0$
- λ_{it} : the shadow value of a unit of account (Euro) at time t
- By the envelope theorem:

$$\frac{dV_i}{dz_0} = \frac{d\mathcal{L}_i}{dz_0}$$

- Following Fagereng et al. (2022) and Del Canto et al. (2023): focus on the "money-metric welfare" $\mathcal{W}_i$:

$$d\mathcal{W}_i = \frac{dV_i/dz_0}{\lambda_{i0}} = \frac{dV_i/dz_0}{u'(c_{i0})} P_{i0}$$

- Divide dV_i/dz_0 by λ_{i0} to get from utils to nominal terms

Welfare Decomposition

$$d\mathcal{W}_i = d\mathcal{W}_i^{DIR} + d\mathcal{W}_i^{UFP} + d\mathcal{W}_i^{IND} + d\mathcal{W}_i^{LR}$$

where welfare change is split into four additive effects:

1. $d\mathcal{W}_i^{DIR}$: short-run pre-government direct effects
 - ▶ Abstracts from all ad-hoc fiscal policies and changes in nominal wages / asset prices
2. $d\mathcal{W}_i^{UFP}$: short-run unconventional fiscal policy component
 - ▶ Incorporates all ad-hoc government responses
3. $d\mathcal{W}_i^{IND}$: short-run indirect equilibrium component
 - ▶ Captures short-run changes in nominal wages and asset prices
4. $d\mathcal{W}_i^{LR}$: long-run component
 - ▶ Captures long-run realignment of relative good prices to the new price level

Welfare 1: Direct Component

$$d\mathcal{W}_i^{DIR} = \left[\underbrace{-\frac{d \log \bar{P}_0^*}{dz_0}}_{\text{average } \pi} - \underbrace{\left(\frac{d \log P_{i0}^*}{dz_0} - \frac{d \log \bar{P}_0^*}{dz_0} \right)}_{\pi \text{ difference (C)}} \right] \times$$

$$\left[\underbrace{W_{i0} - T_{i0}}_{\text{net income (Y)}} + \underbrace{B_{i,S0} + (1 + Q_{L0}\delta)B_{i,L0}}_{\text{net nominal position (NNP)}} + \underbrace{\sum_{k=1}^K D_{k0}a_{i,k0} + \sum_{k=1}^K Q_{k0}(a_{i,0k} - a_{i,1k})}_{\text{dividends + capital gains (K)}} \right]$$

- Measures first-order partial-equilibrium affect, ex-ante with respect to ad-hoc government interventions Empirical
- Substitution effects ignored to first order
- Net nominal position: empirically includes bank deposits and bond holdings net of mortgage and other debt

Welfare 2: Unconventional Fiscal Policy Component

$$d\mathcal{W}_i^{UFP} = - \underbrace{\left(\frac{d \log P_{i0}^*}{dz_0} - \frac{d \log P_{i0}}{dz_0} \right)}_{\pi \text{ difference (C)}} \times$$

$$\left[\underbrace{W_{i0} - T_{i0}}_{\text{net income (Y)}} + \underbrace{B_{i,S0} + (1 + Q_{L0}\delta)B_{i,L0}}_{\text{net nominal position (NNP)}} + \underbrace{\sum_{k=1}^K D_{k0}a_{i,k0} + \sum_{k=1}^K Q_{k0}(a_{i,0k} - a_{i,1k})}_{\text{dividends + capital gains (K)}} \right]$$

$$- \frac{dT_{i0}^{HOC}}{dz_0}$$

- Collect the changes in ad-hoc government interventions specifically implemented in response to the inflation shock
 1. Subsidies and price controls (e.g. in energy markets)
 - ★ Depends on individual share the goods targeted by fiscal intervention
 2. Other ad-hoc transfers (or tax breaks) paid directly to households

Welfare 3: Indirect Equilibrium Component

$$\begin{aligned}
 d\mathcal{W}_i^{IND} = & \underbrace{\frac{d \log W_0}{dz_0} W_0}_{\Delta \text{ wages}} - \underbrace{\frac{d \log T_{i0}^{AUT}}{dz_0} T_{i0}^{AUT}}_{\Delta \text{ net taxes}} \\
 & - \underbrace{\frac{d \log Q_{S0}}{dz_0} Q_{S0} B_{S1} - \frac{d \log Q_{L0}}{dz_0} Q_{L0} (B_{i,L1} - \delta B_{i,L0})}_{\Delta \text{ price of nominal assets}} \\
 & + \underbrace{\sum_{k=1}^K \frac{d \log Q_{k0}}{dz_0} Q_{k0} (a_{i,k0} - a_{i,k1}) + \sum_{k=1}^K \frac{d \log D_{k0}}{dz_0} D_{k0} a_{i,k0}}_{\Delta \text{ price of real assets}}
 \end{aligned}$$

- Includes all equilibrium price changes inducted by the shock
- Expected to offset the impact of the inflation shock at time 0 Empirical
 - ▶ E.g. increase in nominal wage from lagged nominal indexation schemes and contractual renegotiations, increase in the price of real assets

Welfare 4: Long-run Adjustment

$$d\mathcal{W}_i^{LR} = Q_{S0} \left(\frac{d \log P_{i0}}{dz_0} - \frac{d \log \bar{P}_1}{dz_0} \right) \underbrace{[B_{i,S1} + (1 + Q_{L1}\delta)B_{i,L1}]}_{\text{net nominal position at } t = 1}$$

- Reflects the revaluation of net nominal positions at $t = 1$ due to the realignment in relative prices, discounted to time $t = 0$
- Zero only if the shock is neutral across different goods
- If individual i was subject to higher inflation than the mean at $t = 0$, they will see a compensating welfare gain at $t = 1$

⇒ Welfare of the young

$$d\mathcal{W}_i = d\mathcal{W}_i^{DIR} + d\mathcal{W}_i^{UFP} + d\mathcal{W}_i^{IND} + d\mathcal{W}_i^{LR}$$

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Empirical Implementation

- Use several micro datasets to document heterogeneity in key components of the household budget constraint: consumption, income, net taxes, assets and liabilities
- Sort households into 15 groups:
 - ▶ Three ages classes (less than 45, 45-64, older than 64)
 - ▶ Five consumption quintiles within each age class
 - ▶ Use consumption expenditures (on non-durables and services) as a proxy for permanent income, because it is less affected by transitory shocks than current income
- Also use aggregate time series to identify the effects of inflation surprises on asset prices

First Component: Direct Impact of Inflation

$$d\mathcal{W}_i^{DIR} = \left[\underbrace{-\frac{d \log \bar{P}_0^*}{dz_0}}_{\text{average } \pi} - \underbrace{\left(\frac{d \log P_{i0}^*}{dz_0} - \frac{d \log \bar{P}_0^*}{dz_0} \right)}_{\pi \text{ difference (C)}} \right] \times \left[\underbrace{Y}_{\text{net income}} + \underbrace{NNP}_{\text{net nominal position}} + \underbrace{K}_{\text{dividends \& capital gains}} \right]$$

- Household specific inflation rates

- ▶ Inflation rate π_{ic} for household i in country c : $\pi_{ic} = \sum_{j=1}^J xsh_{jic} \pi_{jc}$
- ▶ xsh_{jic} obtained from the latest wave of the Household Budget Survey in 2015 (updated to 2020, assuming relative quantities purchased remained fixed)
- ▶ Good-specific price changes π_{jc} come from the micro data underlying the Harmonized Index of Consumer Prices (HICP)

Equation

Results

First Component: Direct Impact of Inflation

- 20 consumption categories (indexed by j), a refinement of the 12 top-level categories of COICOP
- Do not include imputed rent (in line with HICP but different from US CPI)

Consumption Categories			
Class	Label	Class	Label
01	Food	07.21	Spare parts
02	Alcohol and tobacco	07.22	Fuels
03	Clothing	07.23	Vehicle maintenance
04.1	Actual rent	07.24	Other services for transport equipment
04.3	Dwelling maintenance	07.3	Transport services
04.4	Water supply	08	Communication
04.5	Electricity and gas	09	Recreation
05	Furnishings	10	Education
06	Health	11	Restaurants and Hotels
07.1	Vehicles	12	Miscellaneous

Components of the budget constraint

$$d\mathcal{W}_i^{DIR} = \left[\underbrace{-\frac{d \log \bar{P}_0^*}{dz_0}}_{\text{average } \pi} - \underbrace{\left(\frac{d \log P_{i0}^*}{dz_0} - \frac{d \log \bar{P}_0^*}{dz_0} \right)}_{\pi \text{ difference } (C)} \right] \times \left[\underbrace{Y}_{\text{net income}} + \underbrace{NNP}_{\text{net nominal position}} + \underbrace{K}_{\text{dividends \& capital gains}} \right]$$

- Use the latest wave of the Eurosystem Household Finance and Consumption Survey (HFCS) in 2017
- After-tax income: not directly reported (except for Italy), estimated using data on effective marginal tax rates from OECD as in Slacalek et al. (2020) [Details](#)
- Net nominal position: same as Slacalek et al. (2020) [NNP Details](#)

Components of the budget constraint

- Income flow from real assets (housing and stocks) [Details](#)
 - ▶ Housing rental income reported in HFCS
 - ▶ Stock market dividends using holdings of stock market wealth from HFCS, multiplied by dividend-stock price ratios from Jorda et al (2019)
- Realized capital gains
 - ▶ Construct life-cycle profile of real asset wealth by consumption quintile
 - ▶ Assume households in each age/income bracket plan to attain the same wealth as the immediately older age bracket in the same consumption quintile
 - ▶ Assume all wealth passed on as bequest so the oldest households' portfolio is unchanged

Second Component: Government Interventions

Results

$$d\mathcal{W}_i^{UFP} = \underbrace{\left(\frac{d \log P_{i0}^*}{dz_0} - \frac{d \log P_{i0}}{dz_0} \right)}_{\text{post-tax } \pi \text{ difference}} \times \left[\underbrace{Y}_{\text{net income}} + \underbrace{NNP}_{\text{net nominal position}} + \underbrace{K}_{\text{dividends \& capital gains}} \right] - \underbrace{\frac{dT_{i0}^{HOC}}{dz_0}}_{\text{transfers}}$$

- Fiscal measures that reduced energy prices
 - ▶ E.g., direct subsidies of fuel prices, regulations in electricity markets
 - ▶ These policies affect retail prices directly, so we need to compute counterfactual prices to get from P to P^*
 - ★ Separately for gas, transportation fuel, electricity
 - ★ E.g. Counterfactual gas prices were taken from Dao et al (2023) and assume they apply equally to all four countries
- Government transfers
 - ▶ Incorporate all government interventions that sought to directly compensate each group of households due to the inflation shock (full list in Appendix F)

Third Component: Changes in Wages and Asset Prices

$$d\mathcal{W}_i^{IND} = \Delta \text{wages} - \Delta \text{net taxes} - \Delta \text{price of nominal assets} - \Delta \text{price of real assets}$$

Equation

Results

- Wages and Pension:
 - ▶ Data on negotiated wages by statistical agencies
 - ★ on average wages grew by around 2% in Germany, Spain, and Italy, and around twice as much in France
 - ▶ Take minimum wage into account (all four except Italy)
 - ★ Assume the wage of all working age individuals in the bottom income decile grows at the same rate as the official minimum wage
 - ▶ For over-65: rely on national data on pensions (at least partially indexed in many countries $\implies$ nominal pensions often increased more than wage)

Third Component: Changes in Wages and Asset Prices

● Effects on house prices

[Details](#)

1. Estimate β , the sensitivity of REITs returns to inflation surprises on the days of releases of the German HICP in 2021-2022:
 - ▶ Surprise component measured using inflation swaps
 - ▶ Controlling for Euro area stock returns
2. Estimate δ , the sensitivity of house prices returns $R(H_t)$ to lagged REITs returns
 - ▶ Data from 2006 to 2022, several controls used
3. Back out the estimate of the elasticity of house prices to inflation surprises as
$$\beta \times \delta = -2.042 \times 0.015 = -0.031$$
 - ▶ An inflation surprise of 10% implies a 0.3% drop in house prices

Third Component: Changes in Wages and Asset Prices

- Stock and bond prices
 - ▶ Regress the main stock market index in each country on German HICP inflation surprises, controlling for global stock returns
 - ▶ Similar regression for bond returns (a quantity-weighted average of euro-area government bonds and corporate bonds)
- Elasticities for stock and bond prices are larger than those for house prices: -0.423 and -0.429 , respectively
 - ▶ 10% inflation surprise leads 4.2% decline in stock and bond prices
 - ▶ Stronger reaction and larger volatility of financial asset prices compared to house prices (broadly in line with existing estimates)

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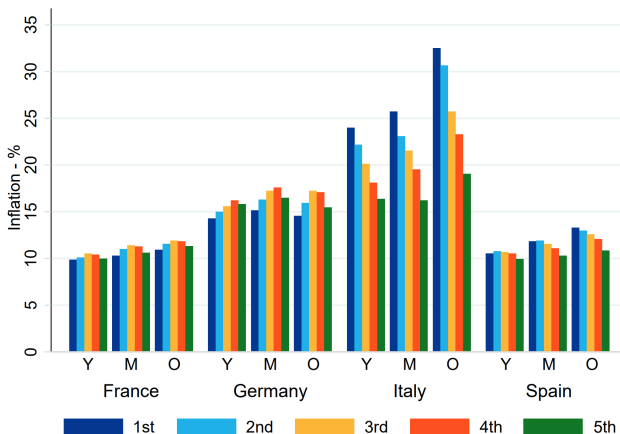
2. Empirical Implementation

3. Results

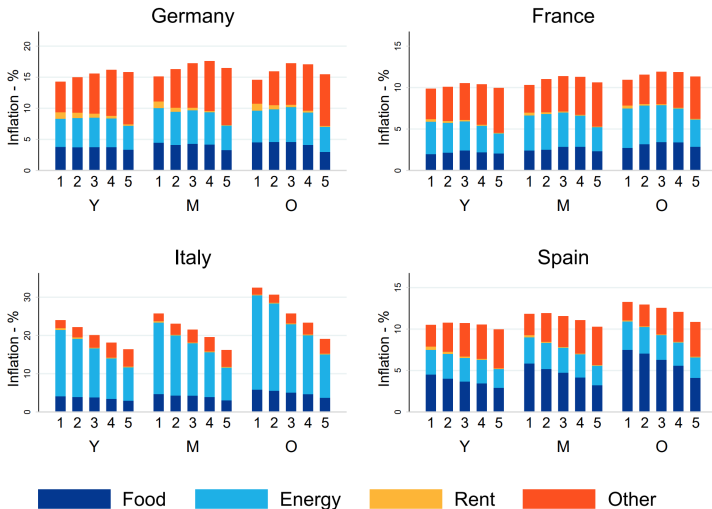
- 3.1. Heterogeneity in Inflation Rates
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Heterogeneity in Inflation Rates

- Country level: 20% in Italy vs. just 10% in France and Spain



Contribution of each category to realized inflation



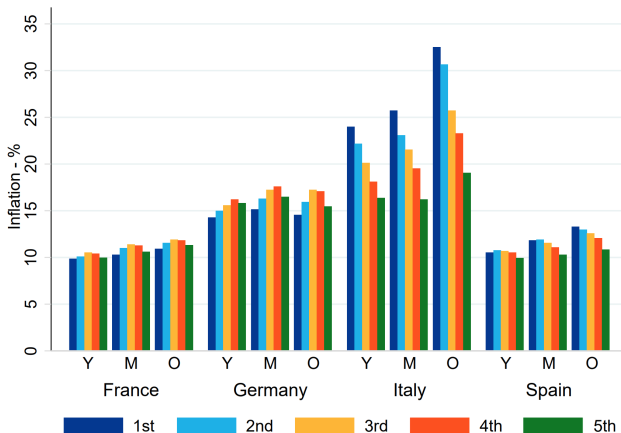
Why would a common energy shock have such different impact across countries?

1. Countries vary in dependence on energy imports (more dependent $\Rightarrow$ larger increase)
2. How energy prices are passed on to consumers depends on market structure and contractual arrangements
 - ▶ Spain: variable pricing in electricity
 - ▶ Germany: long-term contracts renegotiated at an annual frequency
3. Different government interventions

Counterfactual Prices

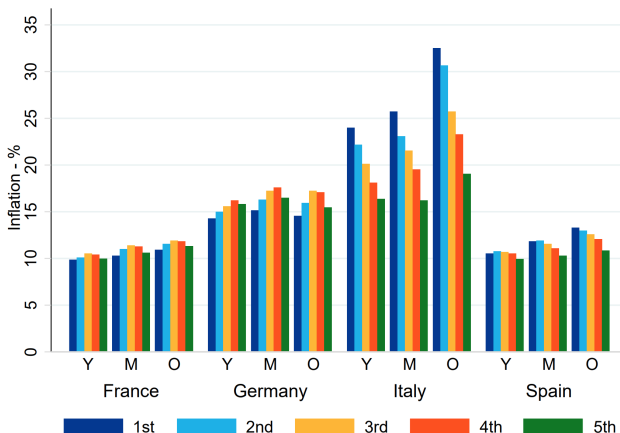
Heterogeneity in Inflation Rates

- Within countries: the inflation rate generally increases with respect to age
 - ▶ Older households spend relatively more in energy and food, but less in rents



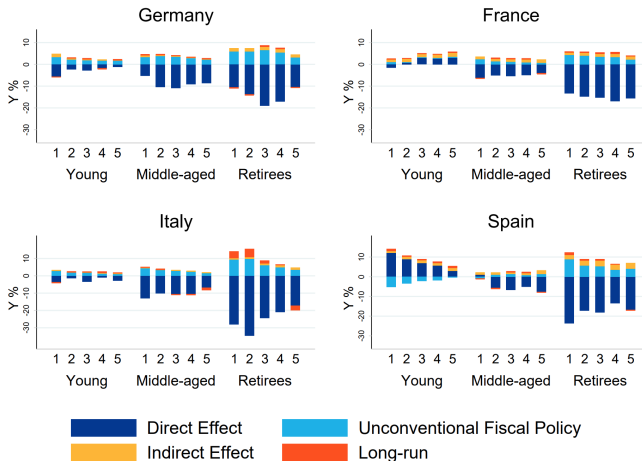
Heterogeneity in Inflation Rates

- Within age groups: the gradient differs by country
 - ▶ Households in lower consumption quintiles spend a higher share of their budget on energy and food (high inflation) as well as rents (sticky)



Decomposition of Total Welfare Effect

- Bulk of welfare impacts (measured as % of biennial disposable income) comes from the direct effect, mitigated by unconventional fiscal policy



1. Direct Effect

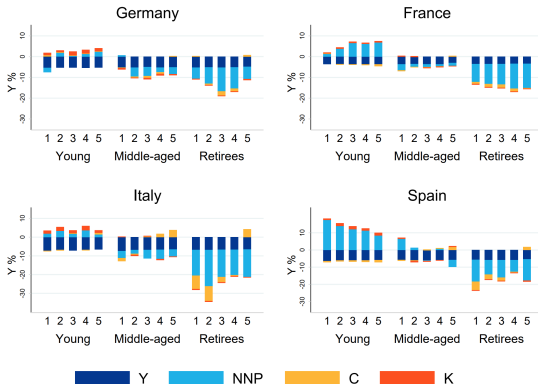
- Considerable heterogeneity
- Losses are markedly larger for Italian households (highest average inflation rate)
- Clear age pattern: old losing more than the young
 - ▶ Particularly striking in Spain and France: the young gains on average



Key Driver of the Age Heterogeneity: NNP Channel

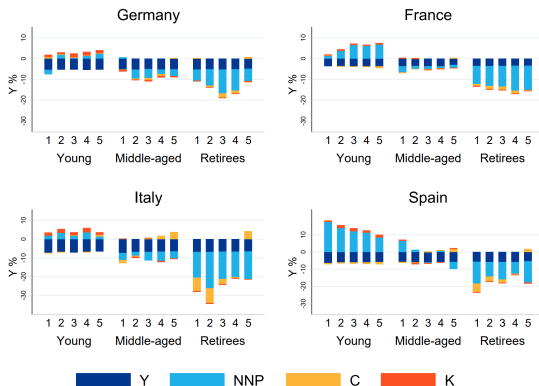
- Old households own more nominal assets (bank deposits or savings accounts)
- Younger households are less likely to have large balances of nominal assets and much more likely to hold nominal debt (mortgages)
- Strong in Spain (relatively high home-ownership rates), much less so in Germany (few households are homeowners with debt)

Measurement



Decomposition of the Direct Effect

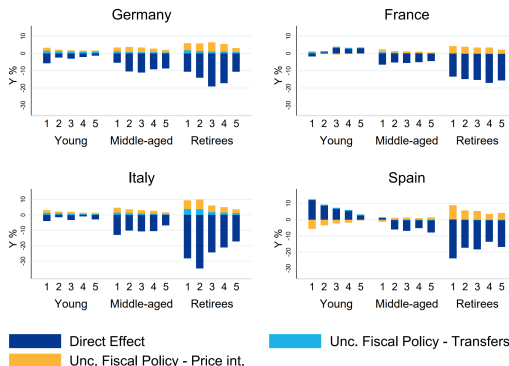
- Net income component (due to wage stickiness) is quite uniform across groups in absolute terms, and plays its biggest role for the middle-aged
- Heterogeneity in consumption baskets across households is only sizable for Italy where inflation is high



2. Unconventional Fiscal Policy

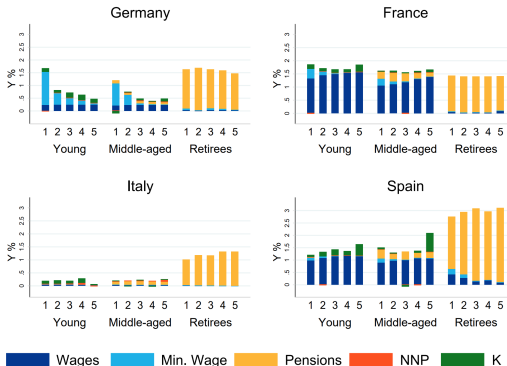
- Substantial effects: reducing the direct welfare effects by 25% for most types
- The role of energy caps was more relevant compared to transfers
- Retirees benefited the most from these measures
 - ▶ Retirees were hard-hit by the shock and has a high share of energy and food consumption (some countries also introduced targeted reliefs)

Measurement



3. Indirect Effect

- Increases in negotiated wages alleviated losses for the young and middle-aged in France and Spain (barely changed for the other two) Measurement
- Rising minimum wages helped the low-income in Germany Magnitude
- Pensions grew the most, benefiting the old



Total Welfare Effect

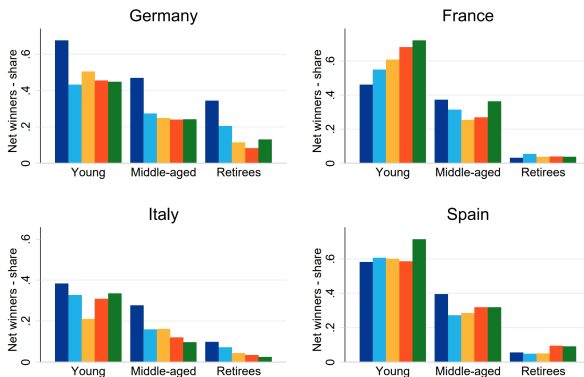
- Young tend to break even in Germany and Italy, and gain about 3-6% of income in France and Spain
- Middle-age tend to lose 3-6%, retirees tend to lose 10% or more

Long run Effects



Share of Net Winners from the Inflation Shock

- On average about 30% of the households gained, and most of them are concentrated in the two younger age groups, who are more likely to have a mortgage
- No clear pattern over the consumption distribution



Conclusion

This paper: measures the heterogeneous welfare effects on households of the recent inflation outburst in the euro area

- Builds a stylized theoretical framework, combined with micro and macro data to construct the empirical exercise
- Finds large welfare effects that are heterogeneous across countries and within countries, across age groups, but not across income groups
 - ▶ 3-4% of disposable income in three countries vs. 8% in Italy
 - ▶ Retirees suffered the largest welfare losses; most net winners are non-retirees
- Direct effects from raw price changes dominates
 - ▶ Key component: erosion of labor income due to wage stickiness and the change in net nominal position
 - ▶ Unconventional fiscal policies also played a non-trivial role in partially shielding households from the shock

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Appendix

Personal Thoughts

- Clever idea, empirics guided by theory
- Evidence and stories seem quite compelling and intuitive
- The model is very general, but only used to do the decomposition
 - ▶ What about market clearing?
 - ▶ How are prices determined in the model? What is the point of OLG?
 - ▶ Explanations in the results sections only based on empirical observations and reasoning (no model prediction)
 - ▶ Where the shock came from is unmodeled
- Envelope theorem and first-order approximations: is this still okay for such a large shock?
- Effects does not depend on consumption quintile (?)
 - ▶ To be verified in the next draft with an additional year of data

Literature Review

- **Methodology:** Applications of the envelope theorem to characterize analytically the first-order impacts of shocks on the macroeconomy (Auclert 2019, Fagereng et al. (2022), DelCanto et al 2023)
- **Empirical channels:** connected to the literature on inflation heterogeneity across household groups (Michael 1979, Kaplan and Schulhofer-Whol 2017, Jaravel 2021, Orchard 2022)
 - ▶ This paper identify differences an order of magnitude larger than those estimated by Hobijn and Lagakos (2005) and Argente and Lee (2020) across the pre-pandemic US
 - ▶ In line with Doepke and Schneider (2006), this paper finds age, not income to be the key dimension due to the strong life cycle profile in NNP
- **Recent Inflation Episode in the euro area:** Most stop at inflation rates
 - ▶ One exception is Cardoso et al. (2022), which examines the relative contributions of the direct effect channels of the current inflation shock in Spain in 2021
 - ▶ This paper: consistent results but also examines explicitly government interventions, assesses four countries in both 2021-2022

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Nature of the Shock

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- Assumption 4: The adjustment in the government budget constraint either occurs through the price level, or through higher real surpluses at $t > 1$
 - ▶ Let B_{Gt} be government debt (a component of household holdings of long-term bonds) with δ_G and price Q_{Gt}
 - ▶ Let S_t be the real budget surplus (deficit if negative): $\bar{P}_t S_t = T_t - \bar{P}_t G_t$
 - ▶ The intertemporal government budget constraint at $t = 0$ is

$$Q_{G0} \delta_G B_{G0} = \bar{P}_0 S_0 + Q_{S0} [\bar{P}_1 S_1 + Q_{G1} B_{G2}]$$

- Two channels:
 1. Through inflation: reducing the value of real debt to equate the present value of all future real surpluses (fiscal theory)
 2. Through future real surpluses (lower spending G_t or less taxes)

The Household Finance and Consumption Survey

- Net Income:

- ▶ Take gross income from HFCS, apply the methodology by Slacalek et al. (2020) to estimate disposable income
- ▶ For France, Germany and Spain, we approximate after-tax income by applying marginal tax rates available from the OECD on taxable income (variable $di1100$) $+ \frac{2}{3} \times$ self-employment income ($di1200$) and adding non-taxable income
- ▶ For Italy, after-tax income is available directly in the HFCS

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The Household Finance and Consumption Survey

● NNP

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- ▶ Following the definition in Doepke and Schneider (2006)
- ▶ Defined as the sum of nominal assets *da2101* (deposits), *da2103* (bonds), *da2107* ("money owed to households") less liabilities *dl1000* ("Total outstanding balance of household's liabilities")
- ▶ The last term includes mortgages and non-mortgage debt (credit lines, credit cards, and other non-collateralized loans)
- ▶ Excludes exposure arising from ownership of shares in financial intermediaries (e.g. mutual funds) or equity

● Housing wealth: *da1110* ("Value of household's main residence") and *da1120* ("Value of other real estate properties")

● Stocks: use directly held stocks reported in *da2105* ("Shares, publicly traded")

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● Rental income: *di1300* ("Rental income from real estate property")

Third Component: Effects on house prices

1. Estimate the sensitivity of REITs returns $R(Q_t)$ to inflation surprises on the days of releases of the German HICP in 2021-2022:

$$R(Q_t) = \beta \Delta ILS_{1Y,t} + \gamma R(S_t) + \epsilon_t$$

where $\Delta ILS_{1Y,t}$ denotes a surprise component of the HICP announcement (measured using swaps) and $R(S_t)$ denotes Euro area stock returns

2. Estimate the sensitivity of house prices returns $R(H_t)$ to lagged REITs returns $R(Q_{t-1})$ using a regression with quarterly data from 2006Q1 to 2022Q2:

$$R(H_t) - R_f = \alpha + \delta[R(Q_{t-1}) - R_f] + \tilde{\gamma}[R(S_t) - R_f] + \text{controls} + \epsilon_t$$

- Controls include broad effective exchange rate, slope of the term structure, growth rate of industrial production, and the growth rate of HICP in the euro area

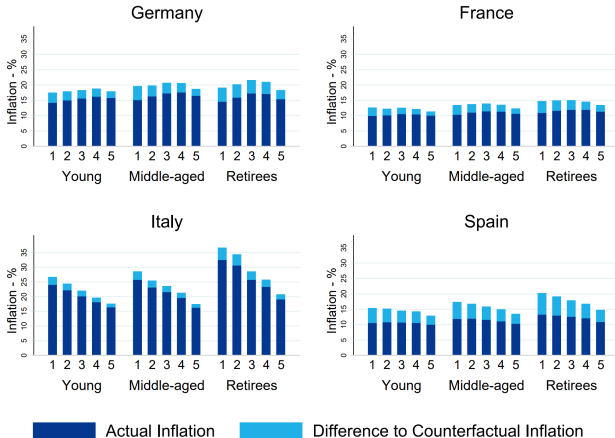
3. Back out the estimate of the elasticity of house prices to inflation surprises as

$$\beta \times \delta = -2.042 \times 0.015 = -0.031$$

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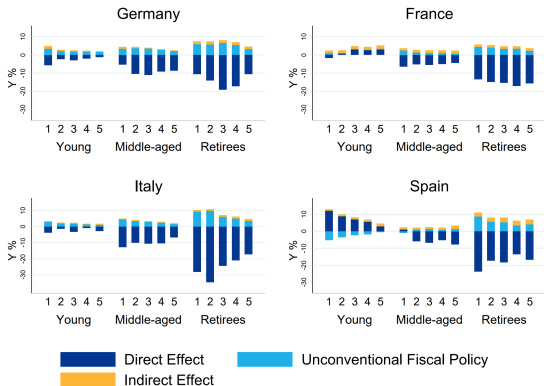
Actual vs Counterfactual Inflation Rates

- Compare with the counterfactual without unconventional fiscal policies (mostly targeted energy prices)

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3. Indirect Effect

- Positive but small, staying in general well below 2% of income

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4. Long Run Effect: Relative Price Adjustment

- Much did not yet materialise during 2021-2022 [Back](#)
- Non-negligible but lower in magnitude than the first two components
- Italy's old (except the richest): experienced losses earlier that will be offset in the long run because of their higher spending share of energy

